DrainSenseAI — Complete Datasets & Sources Reference

Repository: https://github.com/ggthedeveloper/DrainSenseAI | Every link below is clickable. Status reflects code-level provenance verified across ingestion and training scripts.

1. Datasets & Artifacts Committed in the Repository

Dataset / File	Status (Real vs. Synthetic)	Description / Provenance	Downloadable Link
Full Repository Archive DrainSenseAI (main branch)	**Mixed (all committed files)**	Complete GitHub repository ZIP including all committed raw/processed datasets, serialized XGBoost model, and harvesting scripts.	DrainSenseAI-main.zip
data/raw/imd_daily_rainfall_stations.csv	**Real (Manually Curated)**	22 daily station rainfall rows compiled from IMD, APSDMA, and KSNDMC bulletins across VJA, CHE, BOM, and BLR.	imd_daily_rainfall_stations.csv
data/raw/cwc_river_barrage_telemetry.csv	**Real (Manually Curated)**	14 river stage and barrage discharge records compiled from CWC, WRD TN, MCGM, and BBMP reports.	cwc_river_barrage_telemetry.csv
data/raw/isro_bhuvan_flood_inundation_benchmarks.json	**Real (Manually Curated)**	4 city SAR flood benchmark summaries referencing ISRO NRSC Bhuvan (metadata only, not raw 10 m GeoTIFF masks).	isro_bhuvan_flood_inundation_benchmarks.json
data/processed/historical_flood_events.csv	**Real (Manually Curated)**	9 documented historical flood disaster records across Vijayawada, Chennai, Mumbai, and Bengaluru (CSV).	historical_flood_events.csv
data/processed/historical_flood_events.json	**Real (Manually Curated)**	Same 9 documented historical flood disaster events in JSON format.	historical_flood_events.json
data/processed/all_cities_grid_features.csv	**Mixed (Real DEM + Heuristic Proxies)**	5,827 grid rows (BOM: 1,768; BLR: 1,764; CHE: 1,485; VJA: 810). **Real:** Copernicus DEM elevation_m & calculated slope_deg. **Heuristic:** 3-tier concentric distance rings for imperviousness [0.35, 0.65, 0.88], road density [3.8, 8.5, 14.2], building density [0.25, 0.52, 0.78]; simplified polylines & rule-based flood history.	all_cities_grid_features.csv
data/processed/vijayawada_grid_features.csv	**Mixed (Real DEM + Heuristic Proxies)**	810 Vijayawada grid rows (VJA subset of all_cities_grid_features.csv).	vijayawada_grid_features.csv
data/processed/vja_grid_500m.json **data/processed/vijayawada_grid_500m.json**	**Mixed (Real DEM + Heuristic Proxies)**	810-cell 500 m GeoJSON polygon grid for Vijayawada (both files are identical aliases).	vja_grid_500m.json vijayawada_grid_500m.json
data/processed/che_grid_500m.json	**Mixed (Real DEM + Heuristic Proxies)**	1,485-cell 500 m GeoJSON polygon grid for Chennai.	che_grid_500m.json
data/processed/bom_grid_500m.json	**Mixed (Real DEM + Heuristic Proxies)**	1,768-cell 500 m GeoJSON polygon grid for Mumbai.	bom_grid_500m.json
data/processed/blr_grid_500m.json	**Mixed (Real DEM + Heuristic Proxies)**	1,764-cell 500 m GeoJSON polygon grid for Bengaluru.	blr_grid_500m.json

Dataset / File	Status (Real vs. Synthetic)	Description / Provenance	Downloadable Link
data/processed/current_risk_predictions.json	**Model-Generated (Static Simulated Storm)**	Pre-computed VJA 810-cell risk predictions using default static rainfall (24h = 145 mm).	current_risk_predictions.json
data/processed/evaluation_report.json	**Evaluated on Synthetic Labels**	Holdout metrics (N = 155,378; ROC-AUC 0.9823), ROC/PR & calibration curves evaluated against label_generator.py SCS-CN synthetic targets.	evaluation_report.json
models/drainsense_xgb_v1.joblib **models/drainsense_xgb_v1_metadata.json**	**Model Trained on Synthetic Labels**	Calibrated XGBoost classifier and training metadata built from 776,890 rows labeled with SCS-CN + Gaussian noise.	drainsense_xgb_v1.joblib drainsense_xgb_v1_metadata.json
frontend/public/data/ (Primary Cities)	**Model-Generated (Static Simulated Storm)**	Pre-computed UI GeoJSON risk bundles for VJA (810), CHE (1,485), BOM (1,768), and BLR (1,764).	current_risk.json · via · vijayawada che · bom · blr
frontend/public/data/ (Metadata Copies)	**Mixed (UI Copies)**	UI copies of evaluation report, 9 curated historical events, and XGBoost model metadata.	evaluation.json historical_events.json model_metadata.json
data/legacy_prototypes/*_grid_500m.json (8 Retired Cities)	**100% Synthetic (Retired)**	Early Sept 2024 synthetic grids generated via rng.uniform(...) for AMD (1,435), CCU (1,640), COK (1,440), DEL (2,100), GAU (1,215), HYD (1,702), PAT (1,305), PNQ (1,640). Not used by active API.	amd · ccu · cok · del gau · hyd · pat · png
frontend/public/data/current_risk_*.json (8 Legacy Cities)	**Model-Generated on Synthetic Grids**	Pre-computed UI risk bundles generated over the 8 synthetic legacy prototype grids above.	amd · ccu · cok · del gau · hyd · pat · png
docs/assets/*.png (4 README Charts)	**Synthetic Charts**	Illustrative visuals generated by scripts/generate_readme_charts.py using np.random.normal, np.random.exponential, and analytical curves matching summary stats.	catchment_topography_profiles.png rainfall_deluge_hyetographs.png model_performance_matrix.png shap_feature_importance.png

2. API-Fetched & External Datasets (Gitignored or Third-Party)

Dataset / File	Status (Real vs. Synthetic)	Description / Provenance	Downloadable Link
ECMWF ERA5 — Vijayawada (Harvester)	**100% Real**	Hourly precipitation & rain (2024-08-20 to 2024-09-12; 16.5062, 80.6480) used in fetch_real_earth_data.py.	Download VJA ERA5 CSV
ECMWF ERA5 — Chennai (Harvester)	**100% Real**	Hourly precipitation & rain (2023-11-25 to 2023-12-12; 13.0827, 80.2707) for Cyclone Michaung.	Download CHE ERA5 CSV
ECMWF ERA5 — Mumbai (Harvester)	**100% Real**	Hourly precipitation & rain (2023-07-15 to 2023-07-28; 19.0760, 72.8777) for July 2023 Monsoon Surge.	Download BOM ERA5 CSV
ECMWF ERA5 — Bengaluru (Harvester)	**100% Real**	Hourly precipitation & rain (2022-08-28 to 2022-09-08; 12.9716, 77.6200) for Bellandur Cloudburst.	Download BLR ERA5 CSV

Dataset / File	Status (Real vs. Synthetic)	Description / Provenance	Downloadable Link
ECMWF ERA5 — Additional Historic Storms (Primary 4 Cities)	**100% Real**	Additional storm windows defined in rainfall_loader.py & DataHealth.tsx (Asia/Kolkata timezone).	VJA Sep 2024 · VJA Oct 2020 CHE Dec 2023 · CHE Dec 2015 BOM Jul 2023 · BOM Jul 2005 BLR Oct 2024
ECMWF ERA5 — 8 Additional Cities	**100% Real**	Historic flood windows in rainfall_loader.py: DEL (Jul 2023), HYD (Oct 2020), CCU (May 2020), AMD (Jul 2022), PNQ (Sep 2019), COK (Aug 2018), GAU (Jun 2022), PAT (Sep 2019).	DEL CSV · HYD CSV · CCU CSV · AMD CSV PNQ CSV · COK CSV · GAU CSV · PAT CSV
Kaggle Chennai Water Management	**100% Real**	Contains chennai_reservoir_levels.csv & chennai_reservoir_rainfall.csv (84 KB ZIP). Repo links to anuragkanaujia/chennai-water-management and sudalairajkumar/chennai-rainfall-data returned HTTP 404; this is the live working dataset.	Download ZIP (84 KB) Kaggle Dataset Page
Geofabrik India OSM Extract	**100% Real**	Complete India OpenStreetMap PBF extract (~1.71 GB).	india-latest.osm.pbf Geofabrik India Index
Copernicus / SRTM Elevation API (Open-Meteo)	**100% Real**	30 m DEM elevation endpoint queried in 90-point batches by fetch_real_earth_data.py (all 5,827 results are already stored in all_cities_grid_features.csv).	Open-Meteo Elevation Docs Sample VJA Point JSON
OpenStreetMap Overpass API	**100% Real**	Vector waterway query endpoint used via HTTP POST in fetch_real_earth_data.py.	Overpass Turbo · OpenStreetMap
Indian Government & Space Agency Portals	**100% Real (External Portals)**	Official agencies referenced in README, docs, and UI (IMD 2024 PDF timed out when tested).	IMD Mausam · IMD 2024 PDF India-WRIS · NWDP/CWC · ISRO Bhuvan Flood OGD India (data.gov.in) · AP SDMA · TNSDMA · NDMA
Global Satellite, DEM, Land Cover & Hydrology	**100% Real (External Portals / Standards)**	Global remote-sensing repositories and USDA-NRCS SCS-CN engineering methodology cited across docs and UI.	Open-Meteo Weather Docs · Open-Meteo Forecast ESA Copernicus DEM · COP-DEM Data Space Sentinel-1 Browser · NASA GPM IMERG · GES DISC IMERG OpenTopography · USGS EarthExplorer · GEE SRTM ESA WorldCover · USDA-NRCS SCS-CN (NEH-4)

3. Runtime-Generated & Mock Code Fixtures (Not Static Data Files)

Dataset / File	Status (Real vs. Synthetic)	Description / Provenance	Source Script Link
data/processed/drainsense_train_dataset.csv (776,890 rows)	**Mixed Inputs + Synthetic Labels**	Gitignored training matrix cross-joining spatial grids with rainfall windows; binary flood_label is synthesized via SCS-CN runoff + TWI + Gaussian noise. Rebuild locally via python scripts/fetch_real_earth_data.py && python scripts/build_dataset.py.	feature_builder.py label_generator.py

Dataset / File	Status (Real vs. Synthetic)	Description / Provenance	Source Script Link
DRAIN_ASSETS_DATA & ALERTS_DATA	**Static Mock Fixtures**	20 drainage infrastructure assets and 4 incident alerts (dated 2026-09-11) hardcoded as inline dicts in FastAPI endpoints.py with real structure names and simulated siltation/statuses.	endpoints.py
Temporal Nowcast (T+0h..T+6h) & What-If Simulator	**Simulated Scenarios**	Runtime forecasts multiplying base rainfall by fixed multipliers (1.0, 1.35, 1.90, 1.55, 0.85) or user sliders in predictor.py.	predictor.py